

Tableau Dashboard for Performance Analytics and Visualisation

Dynamic Tableau dashboard to visualise the resource usage over sprints

1. Tableau Data source structure

127.0.0.1
MongoDB BI Connector

Database
Osquery_LoadTests_New

Table
MultiCustomer
MultiCustome...onfigdb_data
MultiCustom...ompare_cols
MultiCustome...erge_on_cols
MultiCustome...ightsdb_data
MultiCustom...ompare_cols
MultiCustome...erge_on_cols
MultiCustome...storedb_data
MultiCustom...ompare_cols
MultiCustome...erge_on_cols
MultiCustome...angerdb_data
MultiCustom...ompare_cols
MultiCustome...erge_on_cols
MultiCustome...statedb_data
MultiCustom...ompare_cols
MultiCustome...erge_on_cols
MultiCustome...onfigdb_data
MultiCustom...ompare_cols
MultiCustome...ightsdb_data
MultiCustom...ompare_cols
MultiCustome...angerdb_data
MultiCustom...ompare_cols
MultiCustome...erge_on_cols
MultiCustome...statedb_data
New Custom SQL
New Union
New Table Extension

initial datasource

Connection
Live Extract Edit Refresh
Extract contains all data. 21/04/2025 9:24:09 pm

Filters
0 Add

Load DetailsOverall Usagesavg cpu by each appavg mem by each ...

Overall Usages4 fields 152 rows152 rows

Name
Overall Usages

Fields

Type	Field Name	Physical Table	Remote ...
Abc	_id	MultiCustome...	_id
#	Avg Memory usage (	MultiCustome...	Overall Me...
Abc	node type	MultiCustome...	Overall Me...
#	Avg CPU usage (core	MultiCustome...	Overall CP...

Abc
MultiCustomer_Overall Memory usage
_Id

MultiCustomer_Overall Memory usage
Avg Memory usage (GB)

Abc
MultiCustomer_Overall Memory usage
node type

MultiCustomer_Overall CPU usage by e
Avg CPU usage (cores)

66cae5c4ad59a3ef52bc5557	9.15	airflow	3.300
66cae5c4ad59a3ef52bc5557	3.21	cloudquery	0.690
66cae5c4ad59a3ef52bc5557	755.14	data	140.320
66cae5c4ad59a3ef52bc5557	16.94	ep	1.750
66cae5c4ad59a3ef52bc5557	18.52	pg	25.440
66cae5c4ad59a3ef52bc5557	348.54	process	120.920
66cae7ef97e2efcfef266692	10.95	airflow	3.080
66cae7ef97e2efcfef266692	3.66	cloudquery	1.070
66cae7ef97e2efcfef266692	779.15	data	133.870
66cae7ef97e2efcfef266692	13.97	ep	1.530
66cae7ef97e2efcfef266692	21.01	pg	23.610
66cae7ef97e2efcfef266692	542.25	process	143.940

Data Sourcebar plotscatter plotall time memoryall time CPUavg mem/cpu by each appDashboard 1first chart (2)second chart

initial datasource

Overall Usages is made of 2 tables. ⓘ

MultiCustomer_Overall Memory u...MultiCustomer_Overall CPU usag...

2. Table Relationships

Load Details — Overall Us... ▾

How do relationships differ from joins? [Learn more](#)

Load Details	Operator	Overall Usages
Abc _id (MultiCustomer) ▾	= ▾	Abc _Id ▾
⊕ Add more fields		

Overall Us... — avg cpu by... ▾

How do relationships differ from joins? [Learn more](#)

Overall Usages	Operator	avg cpu by each app
Abc _Id ▾	= ▾	Abc _id (MultiCustomer_res ▾
Abc node type ▾	= ▾	Abc Node type ▾
⊕ Add more fields		

avg cpu by... — avg mem b... ▾

How do relationships differ from joins? [Learn more](#)

avg cpu by each app	Operator	avg mem by each app
Abc _id (MultiCustomer_res ▾	= ▾	Abc _id (MultiCustomer_res ▾
Abc app ▾	= ▾	Abc mem app ▾
Abc Node type ▾	= ▾	Abc Node Type ▾

3. Table schemas

Load Details

7 fields 25 rows

Table Details	Load Details	Load Details	Load Details	Load Details	Load Details	Load Details	Load Details
	_id (MultiCustomer)	Build	Load Duration	Date	Run	Sprint	Stack
	66cae5c4ad59a3ef52bc5557	159051	10	28/07/2024	1	159	S1
	66cae7ef97e2efcfef266692	160042	10	17/08/2024	1	160	S1
	66dfedaa25db27a87a4a6d5d	161013	10	09/09/2024	1	161	S1
	66e03bd824e2ca0acf9aac1a	161049	10	09/09/2024	2	161	Longevity
	66fe9e730319179f84dda377	162021:k8sRuleengine	10	28/09/2024	6	162	S1

avg cpu by each app

4 fields 2039 rows

Table Details	MultiCustomer_resource_usages_analy	Project	Project	Project
	_id (MultiCustomer_resou	app	avg cpu (cores)	Node type
	66cae5c4ad59a3ef52bc5557	.*osqLogger.*	23.920	process
	66cae5c4ad59a3ef52bc5557	.*redis-server.*	0.900	process
	66cae5c4ad59a3ef52bc5557	/lib/systemd/systemd---user	0.230	process
	66cae5c4ad59a3ef52bc5557	/opt/uptycs/cloud/go/bin/ch...	2.520	process
	66cae5c4ad59a3ef52bc5557	/opt/uptycs/cloud/go/bin/cl...	0.490	process

Overall Usages

4 fields 152 rows

Table Details	MultiCustomer_Overall Memory usage	MultiCustomer_Overall Memory usage	MultiCustomer_Overall Memory usage	MultiCustomer_Overall CPU usage by e
	_Id	Avg Memory usage (GB)	node type	Avg CPU usage (cores)
	66cae5c4ad59a3ef52bc5557	9.15	airflow	3.300
	66cae5c4ad59a3ef52bc5557	3.21	cloudquery	0.690
	66cae5c4ad59a3ef52bc5557	755.14	data	140.320
	66cae5c4ad59a3ef52bc5557	16.94	ep	1.750
	66cae5c4ad59a3ef52bc5557	18.52	pg	25.440

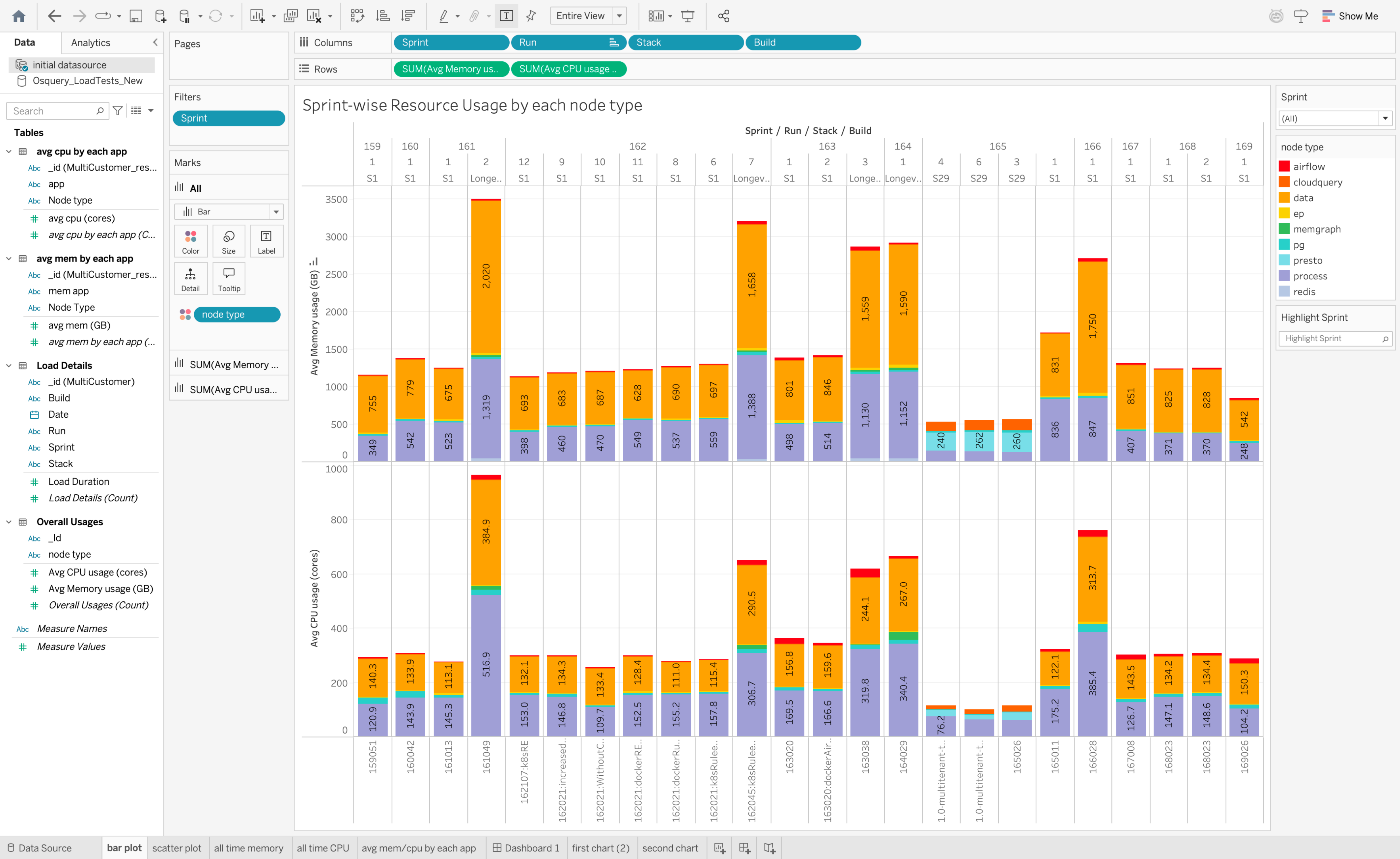
avg mem by each app

4 fields 2081 rows

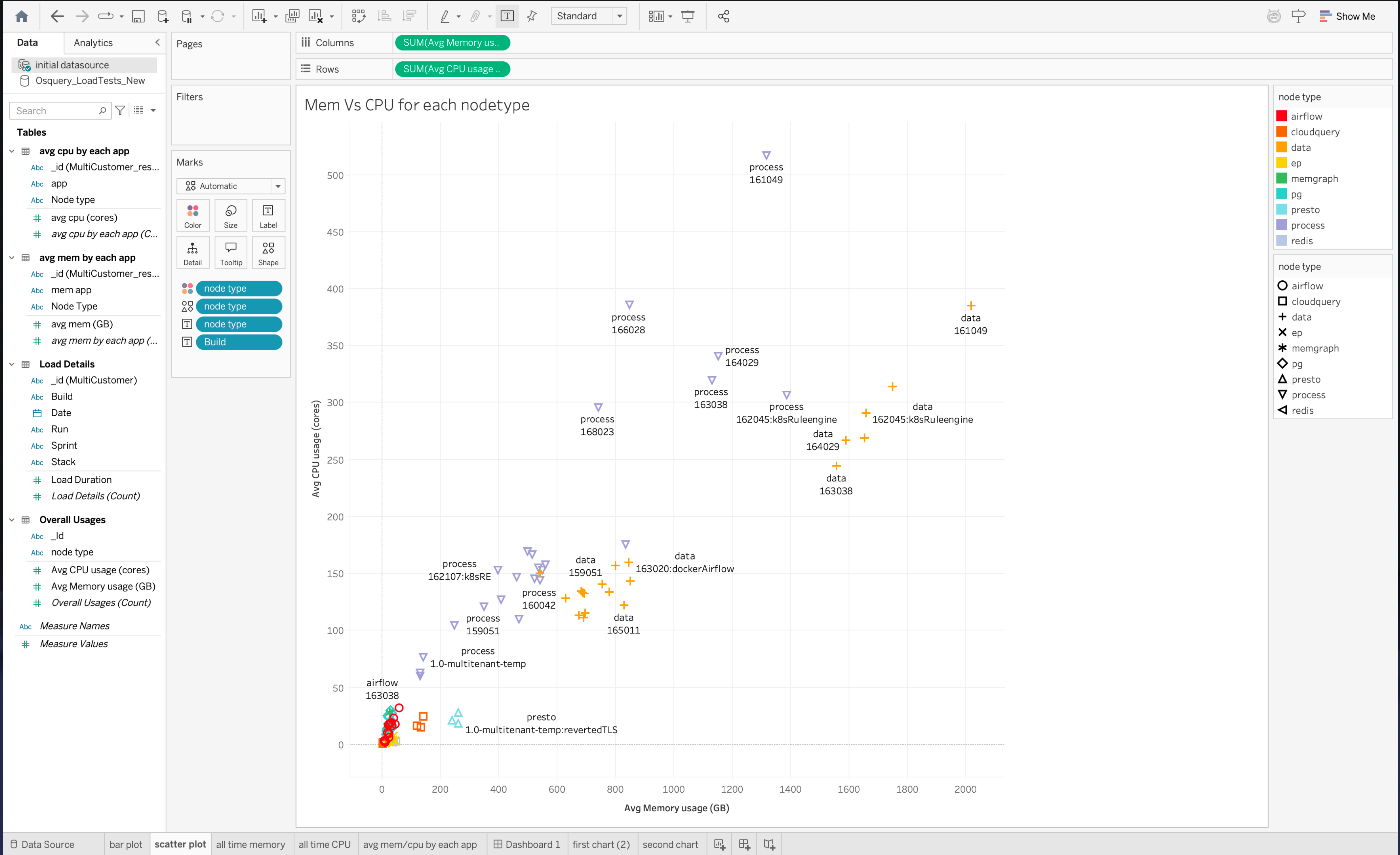
Table Details	MultiCustomer_resource_usages_analy	Project1	Project1	Project1
	_id (MultiCustomer_resou	mem app	avg mem	
	66cae5c4ad59a3ef52bc5557	.*osqLogger.*	27.170	process
	66cae5c4ad59a3ef52bc5557	.*redis-server.*	2.240	process
	66cae5c4ad59a3ef52bc5557	/opt/uptycs/cloud/go/bin/cl...	0.130	process
	66cae5c4ad59a3ef52bc5557	/opt/uptycs/cloud/go/bin/cl...	1.700	process
	66cae5c4ad59a3ef52bc5557	/opt/uptycs/cloud/go/bin/cl...	1.120	process

4. Visualisation Dashboards

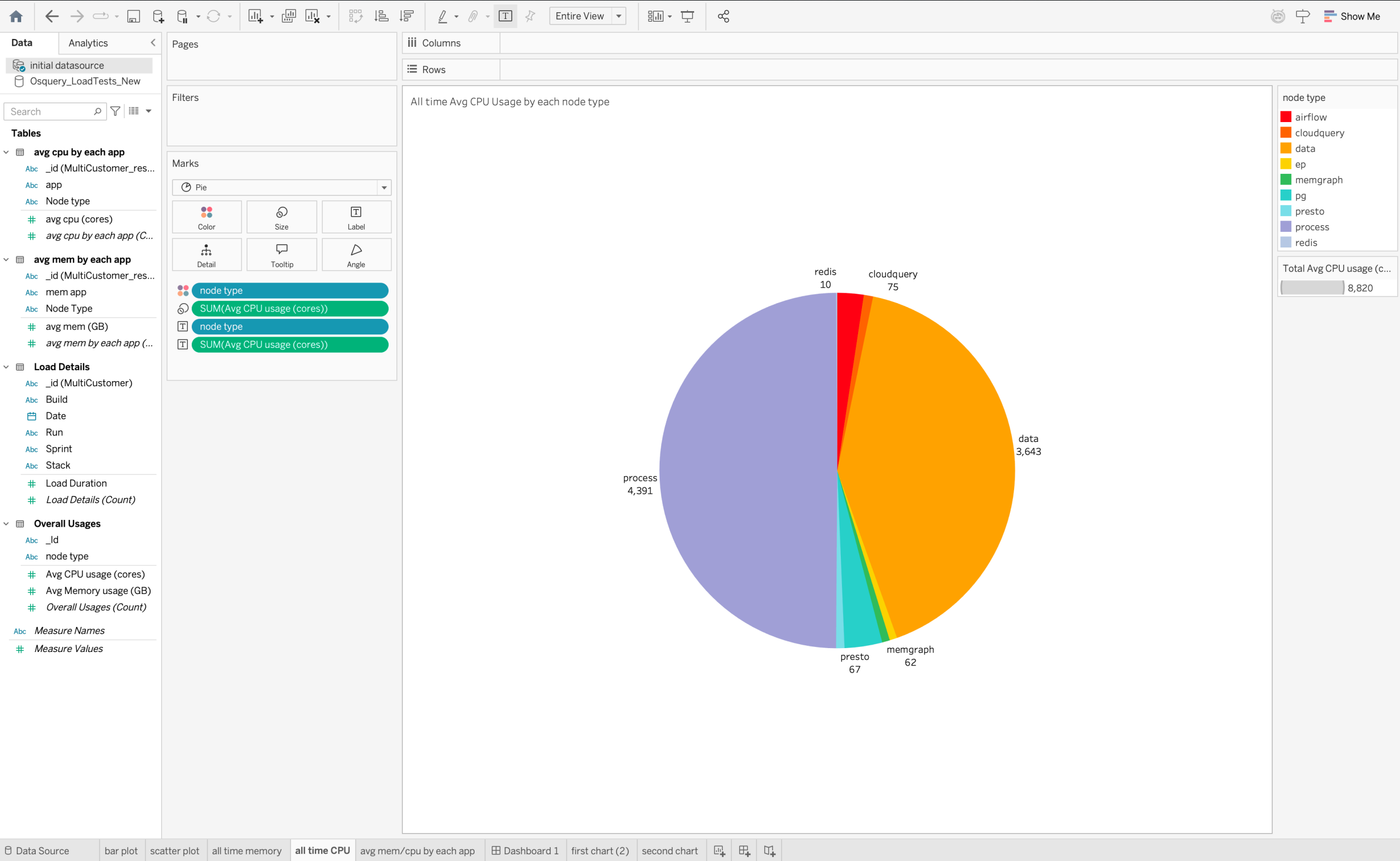
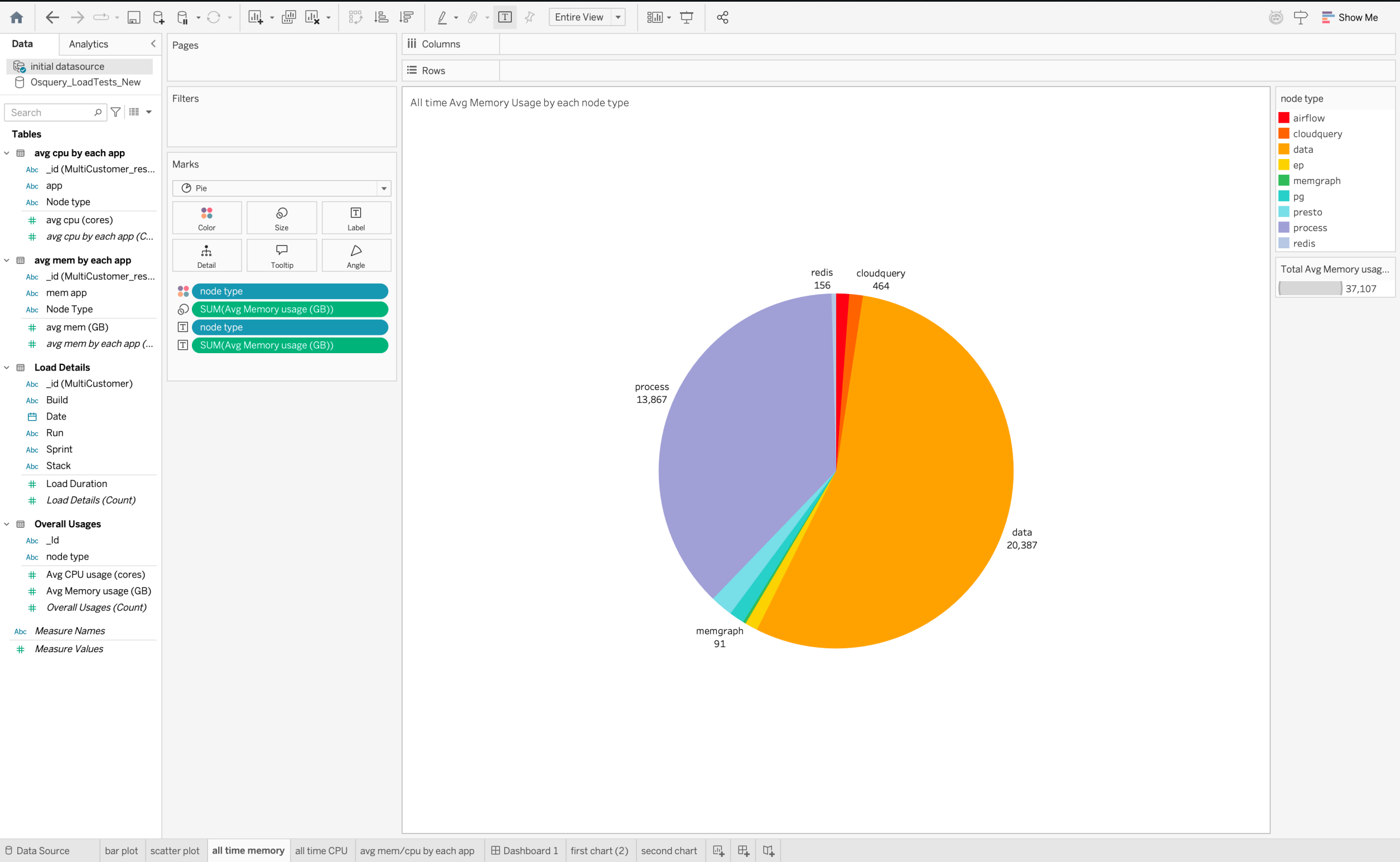
4.1 Bar chart - Shows the Memory,CPU usage analysis for each node-type along with each sprint/build/run and stack.



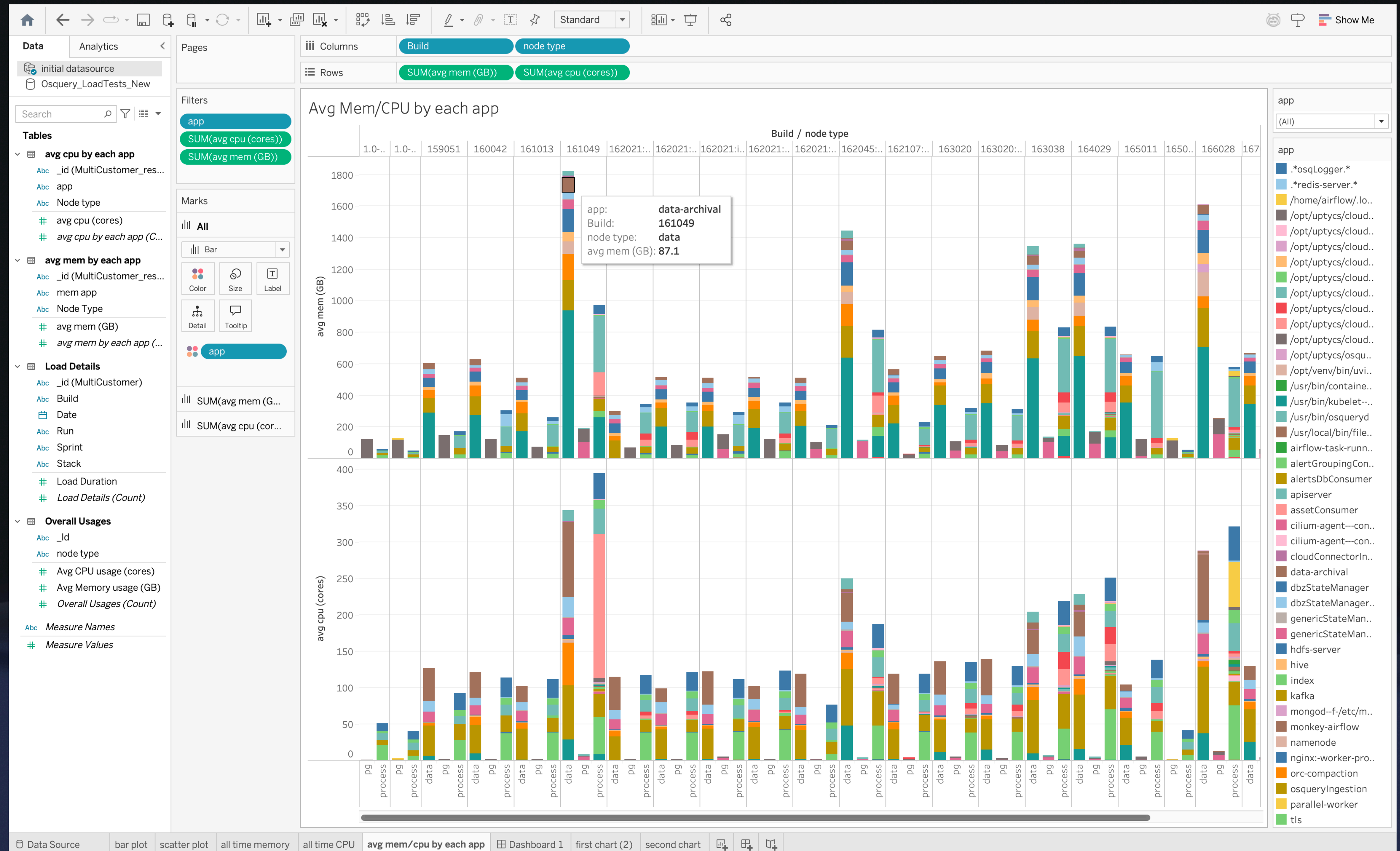
4.2 Scatter plot - Plots the relation between Memory and CPU for each node type



4.3 Pie charts - Plots the contribution of all node types to Overall Memory and Cpu usages



4.4 Stacked bar graph - Shows the distribution of usage by each application for each nodetype divided by build number.



Home page

Report Generator

Home

Dashboards

Load Simulator

Resource Dashboards

Detailed comparison and trend analysis of resource utilization over past sprints

Load Simulator

Manage and Run osquery loads

Generate Performance Load Report

Load Type:

Select Load Type

Load Name:

Select Load Name

Select reports to generate/compare

Sprint Field 1 :

Select Sprint

Run Field 1 :

Select Run

+

Sprint Field 2 :

Select Sprint

Run Field 2 :

Select Run

×

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Enter report title

View Report

Publish to Confluence

Logs

\$ report publishing logs will be displayed here

View Report

Report will be displayed here after clicking 'View Report' Button

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127.0.0.1:8012

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+

Overall cpu usage by each nodetype (cores)

+

Contributors to Resource Usage increase/decrease

+

Complete Analysis piecharts for resource utilizations

+

Memory usages

+

Cpu usages

Osquery - Control plane load Performance Report

178_1 vs 177_1

Load details

Stack : S1

Stack url : jupiter.uptycs.net

Architecture : 1 X 6SU

Sprint : 178

Build : 178090

Load name : ControlPlane

Load type : Osquery

Load duration in hrs : 11.0

Load start time utc : 2024-06-06 16:21

Load end time utc : 2024-06-07 03:21

Load start time ist : 2024-06-06 21:51

Load end time ist : 2024-06-07 08:51

Run : 1

Total number of customers : 1

Test title : Control Plane Load

Total assets : 60K

Redis switchover case time ist :

Error occured while processing Test environment details. Please check if this key has valid format in mongo. 'NoneType' object is not subscriptable

Observations

observation	Status	Comments	
Check for Ingestion lag			

Actions

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Contributors to Resource Usage increase/decrease

Report generated successfully ✕

Memory usages analysis

node_type	Top 5 contributors to increase	Top 5 contributors to decrease
dummy_node_type_0	Dummy_application_0 app : 39.98 GB ⬆️ others (0) app : 0.0 GB ⬆️	Dummy_pod_0 pod : 41.03 GB ⬇️ Dummy_container_0 container : 29.18 GB ⬇️ others (0) container : 0.0 GB ⬇️ others (0) pod : 0.0 GB ⬇️
dummy_node_type_1	Dummy_application_1 app : 29.17 GB ⬆️ others (0) app : 0.0 GB ⬆️	Dummy_container_1 container : 17.61 GB ⬇️ Dummy_pod_1 pod : 12.73 GB ⬇️ others (0) container : 0.0 GB ⬇️ others (0) pod : 0.0 GB ⬇️
dummy_node_type_3	Dummy_application_3 app : 57.47 GB ⬆️ Dummy_container_3 container : 34.91 GB ⬆️ Dummy_pod_3 pod : 34.68 GB ⬆️ others (0) app : 0.0 GB ⬆️ others (0) container : 0.0 GB ⬆️	NaN
dummy_node_type_4	Dummy_pod_4 pod : 92.86 GB ⬆️ Dummy_application_4 app : 47.79 GB ⬆️ Dummy_container_4 container : 9.78 GB ⬆️ others (0) app : 0.0 GB ⬆️ others (0) container : 0.0 GB ⬆️	NaN
dummy_node_type_2	NaN	Dummy_application_2 app : 75.78 GB ⬇️ Dummy_container_2 container : 16.71 GB ⬇️ Dummy_pod_2 pod : 1.26 GB ⬇️ others (0) app : 0.0 GB ⬇️ others (0) container : 0.0 GB ⬇️

Cpu usage analysis

node_type	Top 5 contributors to increase	Top 5 contributors to decrease
dummy_node_type_0	Dummy_application_0 app : 39.98 cores ⬆️ others (0) app : 0.0 cores ⬆️	Dummy_pod_0 pod : 41.03 cores ⬇️ Dummy_container_0 container : 29.18 cores ⬇️ others (0) container : 0.0 cores ⬇️ others (0) pod : 0.0 cores ⬇️
dummy_node_type_1	Dummy_application_1 app : 29.17 cores ⬆️ others (0) app : 0.0 cores ⬆️	Dummy_container_1 container : 17.61 cores ⬇️ Dummy_pod_1 pod : 12.73 cores ⬇️ others (0) container : 0.0 cores ⬇️ others (0) pod : 0.0 cores ⬇️
dummy_node_type_3	Dummy_application_3 app : 57.47 cores ⬆️ Dummy_container_3 container : 34.91 cores ⬆️ Dummy_pod_3 pod : 34.68 cores ⬆️ others (0) app : 0.0 cores ⬆️ others (0) container : 0.0 cores ⬆️	NaN
dummy_node_type_4	Dummy_pod_4 pod : 92.86 cores ⬆️ Dummy_application_4 app : 47.79 cores ⬆️ Dummy_container_4 container : 9.78 cores ⬆️ others (0) app : 0.0 cores ⬆️ others (0) container : 0.0 cores ⬆️	NaN
dummy_node_type_2	NaN	Dummy_application_2 app : 75.78 cores ⬇️ Dummy_container_2 container : 16.71 cores ⬇️ Dummy_pod_2 pod : 1.26 cores ⬇️ others (0) app : 0.0 cores ⬇️ others (0) container : 0.0 cores ⬇️

⚙️ Actions

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Overall memory usage by each nodetype (gb)

Overall cpu usage by each nodetype (cores)

Contributors to Resource Usage increase/decrease

Complete Analysis piecharts for resource utilizations

Memory usages

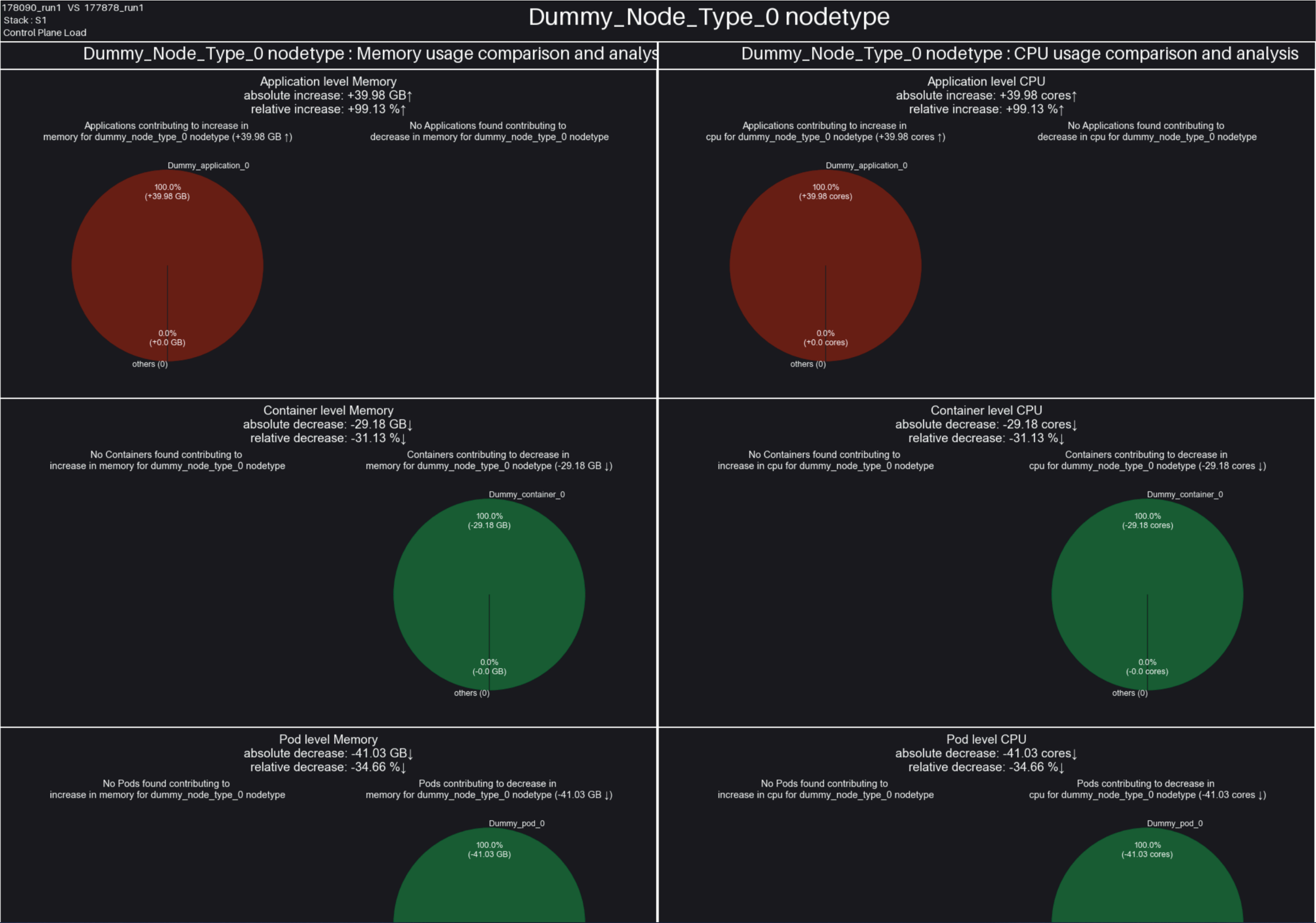
Cpu usages

Complete Analysis piecharts for resource utilizations

dummy_node_type_0

Report generated successfully

Actions



⚙️ Update Simulator Parameters

👤 Load Type:

Stack:

Osquery

s1_nodes.json

👤 Number of Customers to send load to:

👤 Total Number of Assets to Simulate:

100

10000

📊 First X% of customers:

👤 Gets Y% of total load:

Select X = 50

Select Y = 50

🕒 No.of msgs (msgs x 4 = Load Duration in seconds):

📄 Input File:

9000

Select Input File

⌵ Update Params

👁 View Asset Distribution

📄 More about inputfile

Note: Asset distribution will be done only on the selected simulators. So please validate the selected simulators before proceeding.

🟢 1 / 2 Simulators Online

🟡 2 / 2 Simulators Selected

🔄 0 Total Live assets in configdb

✓ Select all

🔄 Pull latest code

📊 CPU / Mem

🔄 Refresh

💀 Kill Instances

👤+ Enroll Assets

▶ Start Load

■ Stop Load

🔄 Online

👤 localhost

🔄 0 Live assets in configdb

0

live instances

100

exp. instances

10000

assets to enroll

227

msgs to send

00:15:08

load duration

🔄 Offline

👤 machine2

🔄 0 Live assets in configdb

Failed to connect to the simulator server
'http://machine2:8123/check_sim_health':
HTTPConnectionPool(host='machine2', port=8123): Max retries
exceeded with url: /check_sim_health (Caused by
NameResolutionError(": Failed to resolve 'machine2' ([Errno 8]
nodename nor servname provided, or not known"))

▼🔧 Update Params

👁 View Asset Distribution

📄 More about inputfile

Note: Asset distribution will be done only on the selected simulators. So please validate the selected simulators before proceeding.

📊 Asset Distribution Logic

Key	Value																					
1.Total assets to enroll	10000																					
2.Total number of customers	100																					
3. Number of simulators configured	2																					
4.Asset Distribution for each customer	<table><tr><th>index</th><th>assets</th></tr><tr><td>0</td><td>100</td></tr><tr><td>1</td><td>100</td></tr><tr><td>2</td><td>100</td></tr><tr><td>3</td><td>100</td></tr><tr><td>4</td><td>100</td></tr><tr><td>5</td><td>100</td></tr></table>	index	assets	0	100	1	100	2	100	3	100	4	100	5	100							
index	assets																					
0	100																					
1	100																					
2	100																					
3	100																					
4	100																					
5	100																					
5.First 50.0% (50) of customers gets	5000 assets.																					
6.And The last 50.0% (50) of customers gets	5000 assets.																					
simulator 'localhost' instance list: (assets=5000)	<table><tr><th>index</th><th>domain</th><th>assets</th></tr><tr><td>0</td><td>jupiter</td><td>100</td></tr><tr><td>1</td><td>jupiter2</td><td>100</td></tr><tr><td>2</td><td>jupiter4</td><td>100</td></tr><tr><td>3</td><td>jupiter6</td><td>100</td></tr><tr><td>4</td><td>jupiter8</td><td>100</td></tr><tr><td>5</td><td>jupiter10</td><td>100</td></tr></table>	index	domain	assets	0	jupiter	100	1	jupiter2	100	2	jupiter4	100	3	jupiter6	100	4	jupiter8	100	5	jupiter10	100
index	domain	assets																				
0	jupiter	100																				
1	jupiter2	100																				
2	jupiter4	100																				
3	jupiter6	100																				
4	jupiter8	100																				
5	jupiter10	100																				
simulator 'machine2' instance list: (assets=5000)	<table><tr><th>index</th><th>domain</th><th>assets</th></tr><tr><td>0</td><td>jupiter1</td><td>100</td></tr><tr><td>1</td><td>jupiter3</td><td>100</td></tr><tr><td>2</td><td>jupiter5</td><td>100</td></tr><tr><td>3</td><td>jupiter7</td><td>100</td></tr><tr><td>4</td><td>jupiter9</td><td>100</td></tr><tr><td>5</td><td>jupiter11</td><td>100</td></tr></table>	index	domain	assets	0	jupiter1	100	1	jupiter3	100	2	jupiter5	100	3	jupiter7	100	4	jupiter9	100	5	jupiter11	100
index	domain	assets																				
0	jupiter1	100																				
1	jupiter3	100																				
2	jupiter5	100																				
3	jupiter7	100																				
4	jupiter9	100																				
5	jupiter11	100																				

Asset distribution logic calculated. ✕